

Analytics Internship Assessment

Case Study 2 – SIOP Reporting

Submitted By

Tanu Chandravanshi

Company

Smith & Nephew Healthcare

Internship Assessment

Analytics Internship Program – 2027 Batch

Submission Month

June 2026

Introduction

As part of the Analytics Internship assessment, I selected Case Study 2 – SIOP Reporting. The objective of this case study was to transform forecast snapshot data, integrate actual sales data, and create an interactive Power BI dashboard for analyzing forecast performance.

The solution was developed to compare different forecast snapshots and provide business users with a clear understanding of forecast accuracy and forecasting trends.

Problem Understanding

In Sales, Inventory and Operations Planning (SIOP), forecasts are generated periodically and stored as monthly snapshots. These snapshots help organizations understand how forecast values change over time.

The main objective of this case study was to:

- Compare multiple forecast snapshots.
- Label the latest snapshot as Resultant.
- Label the previous snapshot as Lag 0.
- Label the earlier snapshot as Lag 1.
- Merge forecast information with actual data.
- Develop an interactive dashboard for business analysis.

Data Preparation and Transformation

The Forecast and Actual datasets were imported into Power BI.

Power Query was used to perform data preparation and transformation activities. During this process:

- Forecast data was prepared and organized.
- Snapshot information was categorized into Resultant, Lag 0, and Lag 1.
- Actual data was merged with forecast data using Item, Country, and Period.
- Data consistency checks were performed before creating the final model.

This approach ensured that the solution remained scalable and easy to maintain.

KPI Calculations

Business metrics were implemented using DAX (Data Analysis Expressions).

Forecast Accuracy % = $1 - \frac{\text{ABS}(\text{Forecast Forecast} - \text{Actual Forecast})}{\text{Actual Forecast}}$

Forecast Bias % = $\frac{\text{Forecast Forecast} - \text{Actual Forecast}}{\text{Actual Forecast}}$

These KPIs help evaluate forecast performance and identify over-forecasting or under-forecasting trends.

Dashboard Development

An interactive Power BI dashboard was created to visualize business performance.

The dashboard contains:

- KPI cards for key business metrics.
- Revenue Forecast vs Actual analysis by Country.
- Non-Revenue Forecast vs Actual analysis by Country.
- CAPEX Forecast vs Actual analysis by Country.
- Lag Type slicer to switch between Resultant, Lag 0, and Lag 1.
- Country slicer for focused analysis.

All visuals dynamically update based on user selections.

Additional Analysis

The developed dashboard provides a clear view of forecast performance and enables comparison across different forecast snapshots.

The solution helps users understand forecast accuracy, identify forecasting bias, and analyze business performance through interactive visualizations and filters.

Overall, the dashboard provides a simple, scalable, and user-friendly approach for supporting data-driven decision-making.

Conclusion

The implementation of this SIOP dashboard provides better visibility into forecast performance and supports more informed business planning. By enabling users to compare current forecasts with previous snapshot lags, the dashboard helps identify forecasting trends, monitor potential forecasting biases, and improve overall decision-making.

Through this case study, I gained valuable practical experience in data transformation using Power Query, implementing DAX calculations, and designing interactive dashboards in Power BI. The final solution provides an efficient, scalable, and user-friendly approach for supporting data-driven business decisions.

Appendix A – Dashboard Preview

